



AI Playbook and Toolkit for Public Health Departments

AI for Case Investigation and Outbreak Response

EPI 300

Case investigation and outbreak response are high-pressure public health workflows that depend on timely information, structured judgment, coordination, documentation, and community trust. AI may support these workflows by summarizing case records, extracting exposures, identifying missing information, prioritizing review, detecting clusters, drafting situation updates, or organizing response tasks. At the same time, errors can affect real people, public health action, and public confidence. This module teaches epidemiologists, informaticians, and response teams how to evaluate and use AI-supported tools in case investigation and outbreak response. The focus is on support, not replacement. AI can help staff work faster and more consistently, but public health judgment, legal authority, and human accountability remain central. Responsible use requires careful attention to data quality, source traceability, sensitivity, uncertainty, equity, escalation, and documentation. AI should help investigators see relevant information more clearly, not hide assumptions or convert uncertain signals into unsupported conclusions.

Level	intermediate/practitioner
Primary track	Epidemiology Role-Based Track
Appears in tracks	epidemiology, operations-and-data-quality, technical-architecture, public-health-executive-leadership

Learning Objectives

- Identify case investigation and outbreak response tasks that AI may support.
- Distinguish between AI-supported triage, summarization, extraction, signal detection, and decision-making.
- Assess data quality, timeliness, source traceability, and equity risks in AI-supported response workflows.
- Define human review, escalation, and documentation expectations for AI outputs.
- Create an AI-supported case investigation or outbreak response workflow review.

Module Overview

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Jurisdiction and Agency Policy Note

This national-level module is intended for training and planning purposes. It does not replace legal, privacy, procurement, civil rights, records, or agency policy review. Learners should confirm applicable requirements in their own jurisdiction and organization before implementing AI-supported workflows.

Public Health Example

During a foodborne illness investigation, AI is used to summarize interview notes and extract possible exposures. Investigators compare the extracted exposures to original notes, review uncertain or conflicting information, and document final interpretations. The tool helps reduce manual review time, but it does not determine the outbreak source or issue public recommendations without epidemiologist review.

Topic Deep Dive

AI can support case investigation by reducing the burden of reviewing large volumes of information. Natural language processing and generative AI may help summarize case notes, identify missing fields, extract dates and exposures, or draft follow-up questions. These tasks are appropriate only when outputs are traceable to source material and investigators can correct errors before the information is used.

Outbreak response may involve signal detection, anomaly review, geospatial patterns, laboratory results, syndromic surveillance, community reports, and partner communications. AI can help identify patterns or organize information, but it can also generate false signals or miss subtle context. A signal should trigger epidemiologic review, not automatic public health action.

Data quality is especially important during response. Early outbreak data are often incomplete, changing, biased toward reported cases, and uneven across communities. AI tools may overstate certainty if they do not represent missingness and uncertainty clearly. Response teams should document what data are preliminary, what

assumptions are being made, and what additional investigation is needed.

Equity considerations should be explicit. Some communities may be less likely to seek care, be tested, respond to interviews, speak English, have stable contact information, or appear in digital datasets. AI-supported prioritization must not systematically deprioritize people or communities because the data available about them are incomplete.

Risks, Failure Modes, and Guardrails

Risks and failure modes include the following:

AI summaries omitting uncertainty or unsupported details.

False outbreak signals creating unnecessary response or public concern.

Missed signals because some communities are underrepresented in data.

Automated prioritization replacing epidemiologic judgment.

Sensitive investigation information being used in unapproved tools.

Recommended guardrails include the following:

Require source traceability for summaries and extracted information.

Use AI outputs as prompts for review, not automatic conclusions.

Document uncertainty, missing data, and preliminary status.

Apply equity review to prioritization and signal detection workflows.

Define escalation to epidemiology leadership for unusual, high-risk, or public-facing outputs.

Reflection Questions

Where does this topic appear in your agency, program, or workflow?

Which staff roles would need to understand or apply this concept before AI use is approved?

What could go wrong if this topic is not addressed before implementation?

What evidence would governance, leadership, or operations staff need before moving forward?

Which policy, data, equity, privacy, security, or workflow questions remain unresolved?

Practical Exercise

Create an AI-supported case investigation or outbreak response workflow review. Include the supported task, data sources, AI output, source traceability, human review point, uncertainty documentation, equity risks, escalation triggers, and monitoring indicators.

Expected Artifact or Evidence

Case investigation or outbreak response workflow review

Source traceability checklist

Uncertainty and missing data documentation

Equity and escalation review

Monitoring indicators

Expected Assignment

- Case investigation or outbreak response workflow review
- Source traceability checklist
- Uncertainty and missing data documentation
- Equity and escalation review
- Monitoring indicators

Knowledge Check

- 1. What should AI outputs do in case investigation?
- 2. What is an outbreak signal?
- 3. Why is uncertainty documentation important during response?
- 4. Which guardrail is appropriate for AI-extracted exposures?
- 5. What is strong completion evidence?

References and Resources

- NIST Artificial Intelligence Risk Management Framework: <https://www.nist.gov/itl/ai-risk-management-framework>
- NIST Generative AI Profile: <https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence>
- CDC Public Health Data Strategy and Data Modernization resources: <https://www.cdc.gov/public-health-data-strategy/php/about/index.html>
- OWASP guidance for LLM application security
- Agency policies for privacy, cybersecurity, records, procurement, accessibility, and public communications
- Relevant public health program SOPs, data governance documentation, and implementation plans